DBMS PROJECT

NexusClubs

A role-based 3NF platform connecting students with club executives and administrators

HTML5 · Tailwind CSS · Oracle Database in 3NF · Node.js Express REST API

PROJECT TEAM

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Problem statement & proposed solution

One consistent platform for students, club leads, and administrators

Students need to

• Find relevant campus events and workshops
• Apply/RSVP and receive a verified digital pass
• Track attendance and view club executives

One shared, consistent 3NF database

All portals read and write the exact same normalized records simultaneously, eliminating data redundancy and state mismatches.

Club Leads need to

• Schedule events without auditorium clashes
• Appoint executive officers and assign custom roles
• Delegate milestone tasks and log roll-call attendance

Role-based portals on one unified 3NF relational database

Student portal

• Browse and search events
• Claim RSVP passes with capacity check
• Track personal passbook status

Shared platform

• Web application (HTML5/Tailwind)
• REST API & in-browser 3NF engine
• Oracle Database in 3NF

Club Lead portal

• Schedule events with clash check
• Appoint officers & assign custom roles
• Delegate tasks and run attendance roll calls

Role-based navigation · Relational 3NF schema (8 tables) · Trigger-based audit trail

Objectives and key features

Objective: a working role-based club management platform with a 3NF database-first design

Secure access

- Student, club lead, and dean personas
- Quick persona switcher for live evaluation
- Navigation rendered per role
- Session state persistence

Student portal

- Search: keyword, venue, host club
- One-click RSVP with unique pass ID
- Auditorium capacity gauges
- Active passbook wallet history

Club Lead portal

- Dashboard metrics (treasury, members, events)
- Appoint officers: Lead gives names and assigns roles
- Schedule events: venue clash check via procedure
- Delegate tasks and log meeting attendance

Technology stack & system architecture

Four layers, each with one job

Presentation · Web UI

HTML5, Tailwind CSS, role-based navigation, responsive modals and glassmorphic panels

Application logic · JavaScript

Modern ES6+, state engine, dynamic executive appointment logic, in-browser 3NF simulator

Data access · Node.js

Express REST API with parameterized routes (/api/events, /api/clubs, /api/rsvp, /api/ledger)

Database · Oracle 3NF

8 normalized 3NF tables, foreign key constraints, 5 analytical views, compound triggers

Architecture flow

ENTRY

index.html
Entry point and navigation router. User role selects the portal.

PAGES (UI)

Student: dashboard · events · my_passes
Lead: dashboard · appoint · tasks · attend
Dean: treasury · audit · collabs

SERVICES

auth_service · event_service
executive_service · attendance_service · ledger_service

DATA ACCESS

REST API (5000)
Parameterized SQL routes
Client 3NF Engine

DATABASE

Oracle Database
8 tables in 3NF · 5 analytical views
constraints · audit triggers

Security: Parameterized SQL queries, foreign key cascading, domain CHECK constraints. Environment: Modern Web Browsers, Node.js Express API (Port 5000), Oracle Database.

End-to-end workflow

From an appointed officer and scheduled event to a verified attendance record

1 · BOTH

Registers or selects persona; role sets the portal

→

2 · CLUB LEAD

Appoints executive officer and assigns custom role

→

3 · CLUB LEAD

Schedules event with venue; trigger checks clash

→

4 · STUDENT

Browses and filters upcoming events

↓

5 · STUDENT

RSVPs and receives unique digital pass token

→

6 · CLUB LEAD

Delegates milestone tasks to appointed officers

→

7 · CLUB LEAD

Opens meeting register and records roll call

→

8 · DATABASE

Procedure and trigger log transaction and audit trail

Key technical implementation

Integrity and business rules are enforced in the 3NF database layer, not only the UI

3NF Normalization

8 normalized tables isolate transitive dependencies and eliminate insertion, update, and deletion anomalies.

Compound triggers

TRG_PREVENT_VENUE_CLASH resolves ORA-04091 mutating table errors by deferring checks to the statement level.

Stored procedures

PROC_SCHEDULE_EVENT and PROC_REGISTER_STUDENT_RSVP guarantee atomic transaction execution.

Budget solvency guard

TRG_BUDGET_GUARD uses FOR UPDATE row locks to prevent overdrawing treasury balance during disbursements.

Audit ledger

Collaboration settlements and grant disbursements are automatically logged in BUDGET_LEDGER.

Integrity constraints

8 tables, foreign keys with CASCADE, CHECK constraints on dates, budgets, and auditorium capacities.

Results, limitations & future scope

What is built and verified so far, and proposed extensions

3

Role-based portals

Student, Club Lead, and Dean with strict role separation

8

Tables in 3NF

Normalized schema with foreign keys, CHECK and UNIQUE domain constraints

100%

Client standalone engine

Plus auto-discovery for live Express REST API on port 5000

Limitations

• Notifications use real-time toast alerts rather than external SMTP email relay.
• Attendance uses digital lead verification rather than hardware RFID badge turnstiles.
• Concurrent bookings use transaction locks but are not stress-tested at high volume.

Future scope · Proposed extensions, not yet implemented

• Mobile QR scanner: verify passes at auditorium doors with automated entry validation.
• Campus ERP integration: direct link with institute MIS database for automated student enrollment and clearance checks.
• Real-time notifications: WebSocket alerts when collaboration proposals or operational tasks are updated.
• Attendance analytics: reporting on member participation rates and quorum consistency trends.

Thank you

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